QUOTA: A Few Configurable Qubits Improve Long-Horizon Rollouts of Physical World Models

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Abstract

World models degrade past the spatial-mean floor over long rollouts, even at foundation scale [Ohana et al., 2024, McCabe et al., 2025]. We hypothesize that the failure sits in the analog part of the signal: physical fields separate into an easily digitized component and a continuous analog component [Mallat, 1989, Daubechies, 1992], so assigning only the analog part to a small configurable quantum register should recover long-horizon accuracy with today's qubit counts. QUOTA (*QUantum Offload of The Analog core*) implements this as one parametrized world model: a Rydberg register whose per-qubit detunings are conditioned on the input state and whose geometry is an optimized blockade-coupling graph, inside a pretrained classical model that carries the rest. Under a pre-registered protocol with parameter- and tuning-matched controls and kill rules on The Well, the register drops into the official CNextU-net checkpoint without harm and, trained through rollouts, is the only arm that improves the hardest long-horizon window on every seed (test VRMSE $\mathbf{0.9528 \pm 0.0542}$, 7.5–7.7% lower than vanilla fine-tuning 1.0301 ± 0.0449 and a matched MLP 1.0322 ± 0.0259; directional at nine trajectories). The trained register is inference-inert yet depends on its interactions during training, accuracy is flat from five qubits upward, and a 3-D MHD grid (three arms, three seeds) is a null; a ten-benchmark industrial ladder spanning fusion, fission proxies, geothermal heat exchange, and video physics extends the protocol. Thirty simulated qubits are classically simulable [Schuld and Killoran, 2022, Cerezo et al., 2025]: the claim is inductive bias on the hardware that exists, not quantum advantage.

1 Introduction

World models — learned simulators that roll a latent state forward under dynamics — increasingly serve as the interface between machine learning and physics [Ohana et al., 2024, McCabe et al., 2025, Gao et al., 2026]. Their central unsolved failure is *physical consistency over long horizons*: on The Well, the best official baseline (CNextU-net) reaches one-step test VRMSE 0.1956 on `turbulent_radiative_layer_2D` (TRL2D) but degrades to 1.01 by rollout window 13:30 — worse than predicting the spatial mean — and even the 1.3B foundation model Walrus exceeds this floor at long horizon [Ohana et al., 2024, McCabe et al., 2025].

Two literatures make claims that bear on this failure. First, *analog quantum simulation*: neutral-atom platforms natively evolve Rydberg Hamiltonians, with inputs encoded directly into detunings, positions, and drives of up to 108 qubits [Kornjača et al., 2024], and Hamiltonian input encoding is provably expressive [Xia et al., 2026]. Second, *quantum modules for foundation models*: unitary and quantum(-inspired) adapters fine-tune frozen large models at extreme parameter efficiency [Koike-Akino et al., 2025, Raj and Coyle, 2025, Aizpurua et al., 2026, Zhao et al., 2026]. To our knowledge,

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neither literature has produced a matched-control win on a public physical-simulation benchmark: quantum-informed priors report turbulence rollout gains, but as offline regularizers against over-parameterized classical priors, with an advantage claim [Wang et al., 2026b, Xue et al., 2026]; the closest in-model attempt, a 5-qubit mixer in a 3D FNO surrogate on proprietary laser-processing data, beat a larger dense-mixer FNO on mean-error metrics but was matched or beaten by its own parameter-matched MLP-mixer control on all but one metric [Papierz et al., 2026]; and reservoir studies show apparent quantum edges evaporating under equal-tuning classical baselines [Pandey, 2026]. Concurrent theory proves that certain classical environments with long-range (irrational-phase) memory admit an exact single-qutrit quantum world model while every finite classical model misaligns the optimal policy [Lumbreras et al., 2026]; a fully observed finite-difference PDE is Markov in its state and lies outside that construction, so our contribution is empirical and control-matched. The similarly named QSWM [Jiang et al., 2026] is quantum-inspired (no register) and, against matched controls on cellular automata, finds its edge confined to short horizons — the opposite profile of ours.

Our premise connects the two claims. Analog quantum hardware may be the natural substrate for physical dynamics, but no practical application fits on today's qubit counts whole — and real simulations are rarely uniformly quantum-hard: they mix classically easy, effectively discrete structure with a harder analog core. Signal analysis offers the analogy: Fourier and multiresolution decompositions separate a field into sparse, effectively discrete structure and continuous spectral content [Mallat, 1989, Daubechies, 1992], which neural operators already exploit classically [Li et al., 2021] (an analogy, not a hardness claim). We therefore adopt the division-of-labor stance that quantum-informed workflows [Wang et al., 2026a] and modular frozen-block designs [Qi et al., 2026] also articulate, but instantiate it in the model rather than in the loss: the qubit budget is a scarce resource to be *allocated*: a reconfigurable, input-parameterized register is steered toward the highest-complexity analog subprocess of the problem at hand, while a pretrained classical world model carries the rest (in the experiments below the classical trunk is co-fine-tuned; the strictly frozen-backbone setting is the archival extension). If the division of labor is right, small fixed budgets become useful on current hardware, and rollout accuracy versus qubit count — "fidelity" throughout means rollout accuracy, not quantum state fidelity — becomes a measurable scaling question rather than a wait for million-qubit machines. The premise is a motivating conjecture: no experiment below identifies or steers the register toward a particular subprocess (it reads a pooled latent), and the alternative that the register acts as an n-independent training perturbation is kept in view. This is the hypothesis our protocol is built to test — and, if warranted, kill.

Contributions:

1. **Checkpoint surgery against a published number.** We insert the bottleneck at the neck of the *official released* CNextU-net checkpoint from The Well and fine-tune, targeting its published 0.1956 one-step test VRMSE [Ohana et al., 2024] with paired per-window Wilcoxon significance over the 873 official test windows — the methodology (official-checkpoint surgery + published-number target + paired statistics) and its result are the contribution, not the architecture [Jain and Kalev, 2026, Hossain et al., 2026, Marcandelli et al., 2025].
2. **An input-conditioned analog ansatz with reconfigurable structure.** Data-dependent detunings, gradient-trained 2D atom geometry, and global drive parameterize the Rydberg generator via exact statevector autodiff; the trained object lies in the native Rydberg Hamiltonian family, while mapping it onto device constraints (sign and static-pattern limits of local detuning on current machines) is future work. ReCon co-trains positions, site couplings, and drive derivative-free at four atoms [DiBrita et al., 2024], ResQ feeds inputs into per-atom detunings on fixed lattices [DiBrita et al., 2025], and PulserDiff differentiates pulses and positions as tooling [Abramavicius et al., 2025]; position gradients were argued ineffective under the divergent r^{-6} term [de Keijzer et al., 2025], which we bound by clamping distances. To our knowledge no prior work composes a learned detuning map with autodiff-trained geometry as a layer of a predictive model (§2).
3. **A pre-registered, kill-rule protocol.** Qubit-count sweeps exist in looser, non-parameter-matched forms [Kim et al., 2025, Kornjača et al., 2024, Vyskubov et al., 2026, Zabihi et al., 2026], and controlled quantum-sidecar insertions with matched bottleneck controls exist for classification [Legziel Levy et al., 2026]; we contribute, to our knowledge, the first *pre-registered, tuning-and-parameter-matched, kill-ruled* instantiation on a public PDE benchmark (significance-tested in the TRL2D campaign, §4.1): matched MLP/product-state con-

trols [Pandey, 2026, Bowles et al., 2024], explicit kill rules (if controls match, the negative result is the paper), and a dequantization suite, under the honest-framing standard of Schuld and Killoran [2022] and the classical-simulability caveat of Cerezo et al. [2025].

4. **A fidelity-versus-qubit-count scaling measurement.** Holding total parameters, data, and step budget fixed, we sweep the register size $n \in \{0, 5, 10, 20, 30\}$ and report rollout accuracy as a function of n, finding it flat within seed noise for $n \geq 5$ (§4.2).

2 System

A classical trunk produces a latent field z; an attention pool yields $h \in \mathbb{R}^d$; a linear map sets per-qubit, per-layer detunings $\delta = \pi \tanh(Wh)$; each 10-qubit register evolves under

$$H_{\mathrm{Ryd}}/\hbar = \sum_i \tfrac{\Omega_\ell}{2} X_i - \sum_i \delta_{\ell,i} n_i + \sum_{i<j} \frac{C_6}{|r_i - r_j|^6}\, n_i n_j$$

with trainable times τ_ℓ, global drives Ω_ℓ, and trainable atom coordinates r_i (geometry-as-parameter); $\langle Z_i \rangle$ readouts re-enter the trunk through a zero-initialized gated residual.

Instantiation. $n_i = |r\rangle\langle r|_i$ is the Rydberg projector on atom i. The evolution is a first-order digital–analog split of H_{Ryd} with L=4 layers: layer ℓ applies, per qubit, the conditioned detuning phase $R_z(\delta_{\ell,i})$ fused with the global drive $R_x(\Omega_\ell)$, then the diagonal interaction phase $\exp(-i\tau_\ell \sum_{i<j} V_{ij} n_i n_j)$ with $V_{ij} = C_6/|r_i - r_j|^6$, $C_6 = 1$ in blockade-radius units and distances clamped at $0.5\, R_b$. Detunings are $\delta_{\ell,i} = b_{\ell,i} + \pi \tanh(W_\ell h)$ with W zero-initialized; Ω_ℓ and τ_ℓ are initialized at 0.5 and 0.3; positions start on a unit lattice with σ=0.05 R_b jitter; readout is $\langle Z_i \rangle$. The block is inserted at the CNextU-net neck (d=672). Each register has 108 Hamiltonian parameters (b, r, Ω, τ) plus the conditioning map W of $80 \times d$ entries (≈2k at d=24 in the arena, 53.8k at the neck); 30 simulated qubits are three *mutually non-interacting* 10-qubit registers (inter-register C_6 omitted; entanglement is intra-register [Huang et al., 2021]). Every baseline implements the identical pool→core→residual interface at matched total parameters (within 0.4% among adapter-bearing arms; the bare trunk sits 0.5% below). Each register trains with a batched complex64 *exact* statevector propagator with autograd gradients, cross-checked against CUDA-Q statevector simulation to $\max_i |\Delta\langle Z_i \rangle| \leq 10^{-6}$.

The register is thus *reconfigurable and input-conditioned*: detunings are functions of the simulation state, and geometry-as-parameter is a natively realizable knob of the platform — optical tweezers assemble atoms into user-defined 2D/3D arrangements at the ∼100-atom scale [Barredo et al., 2018], and coherent arrays now reach 6,100 atoms [Manetsch et al., 2025]; the same Hamiltonian family also supports annealing-style schedules [Ebadi et al., 2022].

Configuring the register. Register configuration is itself a search over parameterized geometric graphs. In this study the configuration is gradient-trained end-to-end and, in one variant, input-conditioned (Table 2); a smoke-scale gradient-free search (CMA-ES, ∼200 evaluations per regime; best VRMSE@8 1.14–1.31) was not competitive with gradient training. Figure 1 shows the learned configurations: a small coherent lattice contraction, with per-input adaptation carried by low-rank detuning modes rather than atom relocation.

3 Protocol

Task and metric. Given a history $(x_{t-3}, \ldots, x_t)$ of field snapshots $x_t \in \mathbb{R}^{C \times H \times W}$, predict x_{t+1}; long-horizon behavior is evaluated by autoregressive rollout; The Well's official rollout scores average VRMSE over step windows 6–12 and 13–30 (denoted 6:12 and 13:30). The official metric is variance-scaled RMSE, $\mathrm{VRMSE} = \|\hat{x} - x\|_2 / \sqrt{\|x - \bar{x}\|_2^2 + \epsilon}$ ($\epsilon = 10^{-7}$; computed per channel and averaged over channels and windows, following The Well's reference implementation), with $\mathrm{VRMSE} = 1$ equivalent to predicting the spatial mean [Ohana et al., 2024]. Throughout, $\pm$ denotes the standard deviation across three seeds.

Hypothesis and kill rule. H1: a quantum bottleneck with input-conditioned Rydberg dynamics improves rollout accuracy over parameter- and tuning-matched classical controls. Kill rule (pre-

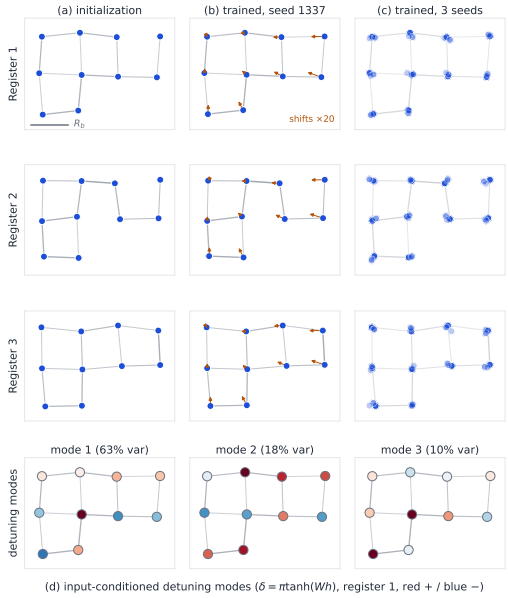


Figure 1: Register configurations (three blockade-decoupled 10-atom Rydberg registers, trainable 2D geometry, blockade-radius units; edges: pairs with C_6/r^6 at least half the coupling at R_b, thicker = stronger). *(a)* Initialization (lattice + σ=0.05 R_b jitter, re-derived from the training seed). *(b)* TD-trained geometry (seed 1337), displacements ×20: a small coherent contraction (mean radial shift $-0.007\ R_b$, all registers and seeds) that strengthens couplings and preserves topology. *(c)* Three seeds agree to $\leq 0.11\ R_b$ rms. *(d)* Per-input reconfiguration via detunings $\delta = \pi\tanh(Wh)$: the top three singular modes of the conditioning map hold 91% of its variance; the dominant mode is shared across registers ($|v_1^{(1)} \cdot v_1^{(2)}| = 0.80$).

registered): if the matched control is statistically indistinguishable from or better than the quantum variant across seeds, the hypothesis is reported dead in this regime. Primary goal: one-step test VRMSE < 0.1956 on TRL2D, significant under a paired per-window Wilcoxon test over the 873 official test windows against the official CNextU-net checkpoint (one-sided, ours $<$ anchor; Holm-corrected across the three seeds; windows overlap within trajectories, so per-window p-values are anti-conservative and are always read together with seed-level consistency). As anchor we re-evaluated the official checkpoint with our pipeline: mean VRMSE **0.1934** over 873 windows (published: 0.1956; 1.1% discrepancy) — we adopt the stricter reproduced number as the bar. Hypotheses, kill rules, and ablations were timestamped in the project repository before the corresponding runs (commit hashes released with the code; self-timestamping is weaker than an external registry). Registration was *sequential*: the one-step endpoint was the original primary; the Round-2 TD protocol — window 13:30, best-checkpoint selection by rollout-validation VRMSE@8, seed-paired comparison against both controls — was specified after Round 1's null was known and fixed before any Round-2 run, so it is confirmatory within its round and sequentially registered relative to the original plan; the interaction ablation and the $\{5, 10, 20\}$q ladder arms were likewise fixed before execution (after the 0q/30q arena runs).

4 Results

4.1 Official benchmark (TRL2D): one-step null, rollout win

The campaign fine-tunes the official released CNextU-net checkpoint under the official protocol, three arms (vanilla continued / matched-MLP control / 30q Rydberg) × three seeds, single-shot test evaluation per best-on-validation model (Round 1: one-step validation VRMSE; Round 2: rollout-validation VRMSE@8, the eighth-step error of an 8-step validation rollout), paired per-window Wilcoxon against the reproduced anchor (Table 1). **Round 1 (one-step objective):** the quantum arm beats the anchor significantly (0.19315 ± 0.00007 vs 0.1934; win rate 52.5–60.1%, $p \leq 1.9\times10^{-2}$ per seed (best 6×10^{-12})) — but vanilla fine-tuning matches it (0.19307 ± 0.00007). All arms

Round	Arm	Test VRMSE (3 seeds)	significance vs anchor	win rate
—	official checkpoint (anchor)	0.1934 / 1.0132	—	—
1 (one-step)	vanilla continued	0.19307 ± 0.00007	3/3 seeds $p < 0.05$	—
1 (one-step)	+ matched MLP (control)	0.19314 ± 0.00016	2/3 seeds $p < 0.05$	—
1 (one-step)	+ 30q Rydberg	0.19315 ± 0.00007	3/3 seeds (best $6{\times}10^{-12}$)	52.5–60.1%
2 (TD), 13:30	vanilla continued	1.0301 ± 0.0449	—	—
2 (TD), 13:30	+ matched MLP (control)	1.0322 ± 0.0259	—	—
2 (TD), 13:30	+ 30q Rydberg	$\mathbf{0.9528 \pm 0.0542}$	seed-level 3/3; per-traj. (n=9) 1/3 Holm-sig. vs MLP, 0/3 vs vanilla	3/3 seeds below anchor

Table 1: TRL2D official-protocol campaign. Round 1: one-step test mean-VRMSE (anchor 0.1934); kill rule fired — attribution null. Round 2: rollout window 13:30 test VRMSE (anchor 1.0132, published 1.01); the quantum arm is the only one below the anchor on 3/3 seeds and beats both controls seed-paired 3/3; per-trajectory Wilcoxon tests (n=9) are underpowered and reach Holm-corrected significance only vs the MLP on one seed (§4.1).

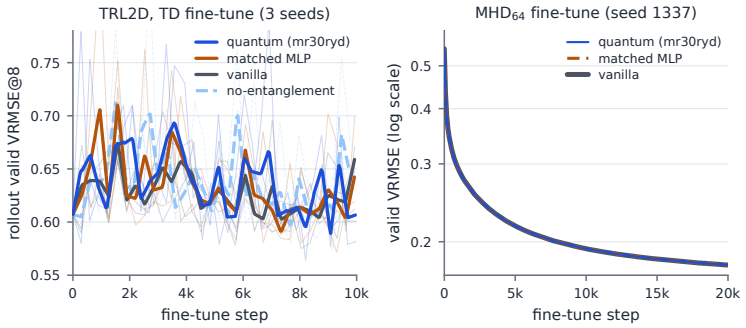


Figure 2: Fine-tuning dynamics. *Left:* TRL2D TD fine-tuning from the official checkpoint (thin: individual seeds; bold: 3-seed mean of the rollout-validation selection metric). Best-checkpoint selection values are quantum 0.564–0.578 across seeds vs vanilla 0.575–0.591, matched-MLP 0.572–0.580, and no-entanglement 0.574–0.592: the arms overlap step-by-step and separate at best-checkpoint selection, matching the directional seed-level evidence of Table 1. *Right:* `MHD_64` fine-tuning (seed 1337, log scale); all three arms converge indistinguishably to validation VRMSE ≈0.177.

improve minutely and indistinguishably: the pre-registered kill rule fired, and we report the attribution null. **Round 2 (temporal-difference/pushforward objective [Brandstetter et al., 2022]: the model is unrolled k=4 autoregressive steps on its own predictions and the loss is applied at the last step, with gradients through that step only),** evaluated on the official test rollout windows (published CNextU-net 0.54 / 1.01 at windows 6:12 / 13:30; our anchor reproduction 0.5482 / 1.0132): the quantum arm is the only configuration that improves the long-horizon window on every seed — 13:30 VRMSE $\mathbf{0.9528 \pm 0.0542}$ vs vanilla 1.0301 ± 0.0449 and matched-MLP 1.0322 ± 0.0259 — winning the seed-paired comparison 3/3 against both controls while matching them on the medium window (6:12: quantum 0.5278, vanilla 0.5216, MLP 0.5271, anchor 0.5482). At the trajectory level the test set is small (n=9 test trajectories), and paired per-trajectory Wilcoxon tests on window 13:30 are correspondingly underpowered: the quantum arm beats vanilla on 6/9, 7/9, and 7/9 trajectories across seeds (one-sided $p = 0.21, 0.037, 0.027$; none survives Holm), beats the matched MLP on 5/9, 6/9, 7/9 ($p = 0.50, 0.21, 0.0098$; the last survives Holm), and beats the anchor on 4/9, 5/9, 6/9 ($p = 0.54, 0.33, 0.065$) — so the seed-mean improvement is carried by a subset of trajectories. This Round-2 evidence is therefore directional seed-level consistency, not trajectory-level significance under the pre-registered standard. TD arms are rollout models: their one-step test error rises by design (to 0.31–0.33 across all three arms, vs 0.193 for the one-step arms); Figure 2 (left) shows the fine-tuning trajectories — arms overlap step-by-step and separate at best-checkpoint selection, consistent with the seed-level framing. Figure 4 visualizes a full rollout: prediction error is an *interface* phenomenon, concentrated where the density gradient is largest, and the quantum arm is the best of the TD arms at tracking the front's position.

4.2 Qubit-count scaling

The premise predicts that, at fixed total parameters and budget, rollout fidelity should improve as the analog register grows. We measure this in an internal fixed-budget *sprint arena* (a 3-regime TRL2D

Register (fixed total params)	Qubits	VRMSE@8 (3 seeds)
FNO trunk alone	0	0.9567 ± 0.2554
+ Rydberg register	5	0.8427 ± 0.0709
+ Rydberg register	10	0.9332 ± 0.2166
+ Rydberg registers	20	0.9269 ± 0.2792
+ Rydberg registers (3×10)	30	0.8447 ± 0.0511
+ matched MLP 30-dim (control)	—	0.9534 ± 0.1354
+ 30q, conditioned geometry	30	0.8951 ± 0.1367

Table 2: Rollout accuracy versus qubit count (internal sprint arena, fixed budget, 3 seeds; $\pm$ = sample SD). The 0q and 30q arms ran first (arena study); the interior $\{5, 10, 20\}$q arms were pre-registered before execution and completed before submission. Points above 10 qubits add registers of 10, not larger registers. The curve is flat within seed noise for $n \geq 5$: the ladder bounds, not resolves, any rise ($\mathrm{VRMSE@8}(n) = 0.849 + 0.108\, e^{-n/n^\star}$ with fitted $n^\star \approx 0.1 \ll 5$: saturation below the smallest tested register, and a constant fits the $n \geq 5$ points equally well).

subset, fixed 2,000 steps, TD k=4, 3 seeds; the metric, VRMSE@8, is the VRMSE of the eighth autoregressive rollout step averaged over strided starts; ∼1.33M total parameters, matched within 0.4% across variants; internally comparable, *not* comparable to the official table). The training objective interacts strongly with the register: switching from one-step to TD training improves the quantum variant by 23% ($1.1002 \pm 0.1939 \rightarrow 0.8447 \pm 0.0511$), the classical control by 10% ($1.0574 \pm 0.0574 \rightarrow 0.9534 \pm 0.1354$), and the bare trunk by under 1% ($0.9637 \pm 0.0048 \rightarrow 0.9567 \pm 0.2554$) — in this arena, adapters only show their edge under rollout-aware training. Under TD, the 30-qubit Rydberg bottleneck beats the matched-MLP control on 3/3 seeds (0.903/0.826/0.806 vs 1.098/0.933/0.829) with the best mean and lowest variance; an input-conditioned-geometry variant reaches 0.8951 ± 0.1367 — worse than the static-layout register, consistent with adaptation being carried by detunings on this task. Table 2 and Figure 3 report the completed ladder. The measured answer is *immediate saturation*: the five-qubit register already reaches the 30-qubit level (0.8427 ± 0.0709 vs 0.8447 ± 0.0511), intermediate counts are statistically indistinguishable within seed noise (10q 0.9332 ± 0.2166; 20q 0.9269 ± 0.2792), and the three-constant saturating fit degenerates to a step at the smallest tested register — $\mathrm{VRMSE@8}(n) = 0.849 + 0.108\, e^{-n/n^\star}$ with fitted $n^\star \approx 0.1 \ll 5$: saturation below the smallest tested register, and a constant fits the $n \geq 5$ points equally well. Given the 0-qubit arm's seed variance (the 5-qubit register is better on 2/3 seeds), even the step is at the edge of resolution: the ladder *bounds* where the curve could rise rather than establishing a law. Two readings are compatible with a flat curve: that this arena's analog core is small enough for five qubits (the premise's reading), or that the register acts as an n-independent perturbation of TD training — the reading a frozen-block bound predicts [Qi et al., 2026]. One datum separates them only weakly: the 30-dimensional matched MLP (0.9534 ± 0.1354) does not reach what five qubits reach (0.8427 ± 0.0709), so the *structure* of the bottleneck rather than its width appears to matter — suggestive at three seeds, not conclusive.

4.3 Mechanism: inference-inert registers whose interactions matter in training

Our dequantization suite shows the trained adapter is *inference-inert*: product-state and χ=1 surrogates, 128-shot readout, and even zeroing the readout reproduce the fine-tuned model's rollout VRMSE to within 2×10^{-5} (600× below seed noise). The mean von Neumann entropy across the balanced 5|5 bipartition of each trained 10-qubit register, over validation inputs, is 0.03–0.09 bits; that zeroing the readout changes nothing means the trained gate is effectively closed. The quantum arm's gain therefore points to a *training-dynamics* effect — the bottleneck steers TD fine-tuning into a better basin that is then encoded in the trunk — consistent with our pre-registered framing of quantum-structured inductive bias rather than inference-time quantum computation. Neighbors report pieces of this pattern in other settings — faster convergence of hybrid PINNs [Klement et al., 2026], quantum-assisted training with classical inference by design [Chen, 2026], and reservoirs whose short-range entanglement is load-bearing at inference yet MPS-simulable [De Lorenzis et al., 2025] — but none tests both inference-inertness and training-side interaction necessity. In the pre-registered training-side ablation, retraining the identical adapter with blockade interactions zeroed — product state from step 0, same TD protocol, three seeds — loses the effect: 13:30 test VRMSE 1.0156 ± 0.0433 with only 1/3 seeds below the anchor (vs 3/3 entangled), and the entangled arm's

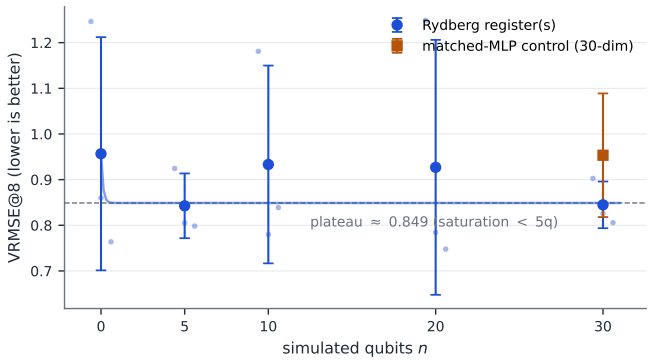


Figure 3: Rollout error versus qubit count at fixed total parameters, data, and step budget (3 seeds, mean ± SD; amber: matched-MLP control), with the degenerate saturating fit $\mathrm{VRMSE@8}(n) = 0.849 + 0.108\, e^{-n/n^\star}$ with fitted $n^\star \approx 0.1 \ll 5$: saturation below the smallest tested register, and a constant fits the $n \geq 5$ points equally well.

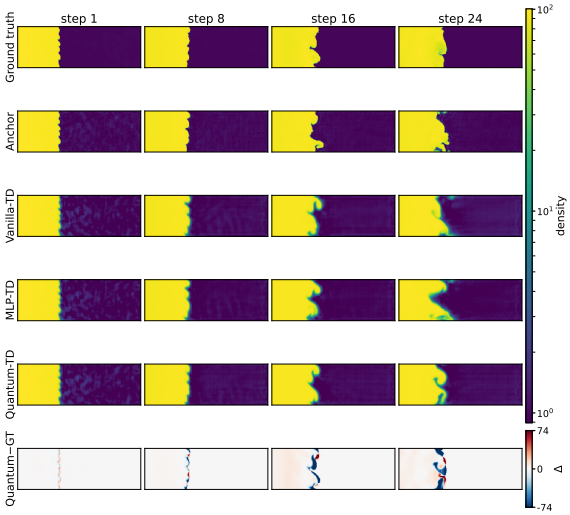


Figure 4: Ground truth vs model rollouts on the first TRL2D test trajectory (density, log scale; hardest cooling-time regime; rows: models, columns: rollout steps; bottom: signed Quantum-TD error). Error concentrates at the mixing front: at step 24, 83% of Quantum-TD squared density error lies in the top-decile $|\nabla\rho|$ band (10% of pixels). On this one trajectory the anchor has the lowest late-step VRMSE (0.640/0.659 at steps 16/24 vs Quantum-TD 0.890/0.712, best TD arm) while Quantum-TD tracks the front best (mean displacement 8.7 px vs 17.0 vanilla, 22.1 MLP, 10.0 anchor); headline claims rest on full-test-set averages (Table 1).

rollout-validation selection is better on every seed (0.5783/0.5771/0.5642 vs 0.5923/0.5807/0.5739). Zeroing V_{ij} removes entanglement *and* the mean-field nonlinearity of the interaction term, so the licensed conclusion is that the blockade interactions appear dispensable at inference yet necessary during optimization in this regime; separating entanglement from mean-field interaction requires a Gutzwiller-type control (interactions kept, entanglement forbidden). A pre-registered mean-field pilot in the arena — a product state by construction — matches the entangled register on all three seeds (VRMSE@8 0.839 ± 0.046 vs 0.845 ± 0.051; 0.891/0.825/0.802 vs 0.903/0.826/0.806), pointing to the interaction nonlinearity rather than entanglement; the decisive official-protocol arms complete after submission. The gain tracks training *through* interacting Rydberg dynamics rather than a generic high-dimensional bottleneck.

4.4 Cross-domain preliminary: MHD_64

On The Well's 3-D, 7-channel `MHD_64`, the released official CNextU-net checkpoint does not reproduce its paper claim of 0.1633 under our evaluation pipeline — the same pipeline reproduces the TRL2D anchor to 1.1%, which localizes the discrepancy to the released artifact (reported upstream): we measure ≈0.66 on both validation and test. Our checkpoint-surgery fine-tunes recover it to the 0.177 level (single seed, validation only, test untouched): at 20k steps all three arms — vanilla, quantum 3×10q, and matched-MLP — reach validation VRMSE 0.1770–0.1771 (undifferentiated at one seed; Figure 2, right). This is preliminary cross-domain evidence that the surgery pipeline transfers; no quantum-versus-control claim is made at one seed.

5 Limitations and future work

This is a preliminary study: three seeds per arm, one primary benchmark, a single-seed validation-only MHD probe, an internal sprint arena whose ladder saturates at its difficulty, and per-trajectory tests underpowered at nine trajectories. All quantum components are classically simulated; at $\leq$30 qubits every claim is an inductive-bias claim, not a quantum-advantage claim [Schuld and Killoran, 2022, Cerezo et al., 2025], the stance is utilitarian, and the trunk sees only n linear readouts per register. Every experiment, including each negative result, runs on one 121 GiB workstation and regenerates from one command per table.

Positions on the workshop's open questions. *(1) Correctness:* physical consistency over official rollout windows against the spatial-mean floor, paired per-window tests, pre-registered kill rules, and diagnosis of where error lives (Fig. 4) — not pixel fidelity alone. *(2) Pixels vs. latents:* the surrogate predicts in observation space (inspectable rollouts) while the register acts on a pooled latent and returns n readouts, an identifiable low-dimensional latent by construction. *(3) Long horizons:* the only gain appears on the hardest horizon window and only under rollout-aware training; action-controllability is untested. *(4) Sim-to-real:* the simulation-trained surrogate still degrades past the spatial-mean floor at long horizons, so it augments rather than replaces the solver; device mapping of the register is untested. *(5) Scaling laws:* accuracy versus qubit count is flat from five qubits at this difficulty; the question must be posed on harder analog cores.

Future work. (i) *Power:* the full 3×3 MHD grid, more seeds and trajectories. (ii) *Mechanism:* the official-protocol Gutzwiller arms and an entanglement-versus-gain correlation. (iii) *Harder analog cores:* the ladder, extended above 30 qubits by register count and size, on 3-D MHD, frozen video world models scored by physics judges [Gao et al., 2026, Team Wan et al., 2025, Motamed et al., 2025, Li et al., 2025], and industrial energy surrogates (tokamak transport and magnetics, equilibrium reconstruction, geothermal plumes). (iv) *Configuration search:* an evolved register bank with routing, reinforcement-learning and quantum-architecture-search over parameterized register graphs [Zoph and Le, 2017, Du et al., 2022], and temporal-difference test-time adaptation. (v) *Hardware:* device-constraint mapping, emulator fidelity, small hardware micro-runs.

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